

BTS SIO - Option SISR

PROCEDURE TECHNIQUE

TP 2 - Installation d'un Serveur LAMP Securise

Module	Administration de Serveurs Linux
TP N°	02
Titre	Installation et securisation d'un serveur LAMP
Niveau	BTS SIO - SISR 1ere annee
Duree estimee	4 heures
Environnement	Ubuntu Server 22.04 LTS
Competences	B1 - Installer / B4 - Securiser

1. Objectifs du TP

L'acronyme LAMP designe Linux, Apache, MySQL/MariaDB, PHP. Ce TP couvre l'installation complete et la securisation de cette pile. A l'issue du TP, vous serez capable de :

- Installer et configurer Apache2
- Installer et securiser MariaDB
- Installer PHP et le connecter a Apache et MariaDB
- Configurer le pare-feu UFW
- Mettre en place HTTPS avec un certificat auto-signé

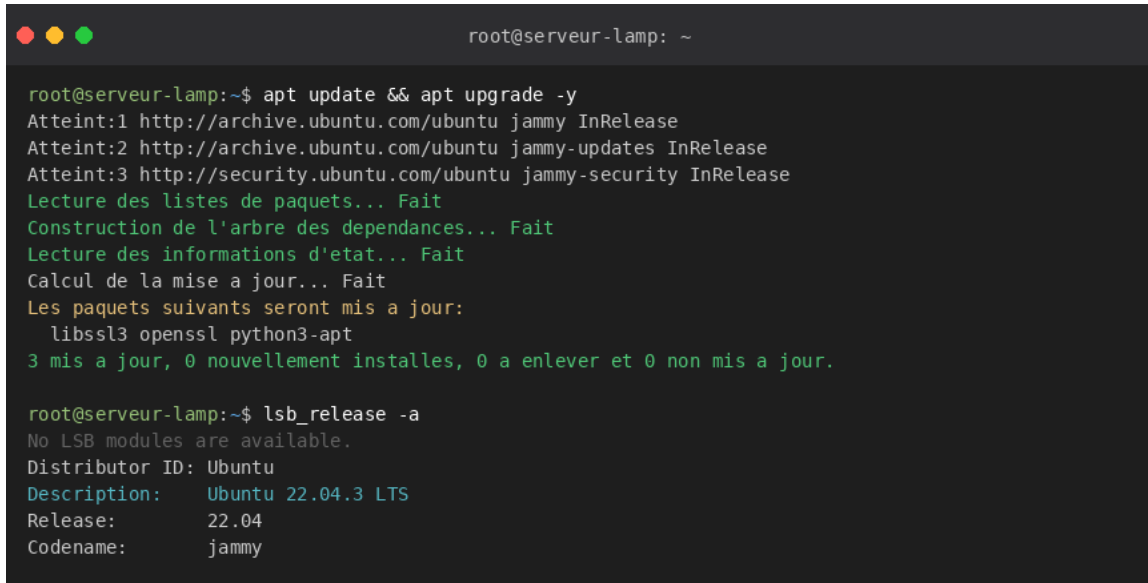
2. Prerequis

Element	Detail
OS	Ubuntu Server 22.04 LTS (minimal)
RAM	Minimum 1 Go (2 Go recommandes)
Disque	Minimum 10 Go libres
Reseau	Acces Internet + IP statique
Acces	Compte avec privileges sudo

ATTENTION : Toutes les commandes doivent etre executees avec les droits sudo, pas directement en root.

3. Preparation du systeme

```
sudo apt update && sudo apt upgrade -y
lsb_release -a
sudo hostnamectl set-hostname serveur-lamp
```

A terminal window titled 'root@serveur-lamp: ~' showing the execution of system update and lsb_release commands. The first command, 'apt update && apt upgrade -y', outputs information about updates from archive.ubuntu.com and security.ubuntu.com, lists packages to be updated (libssl3, openssl, python3-apt), and states that 3 packages will be updated. The second command, 'lsb_release -a', outputs system information: Ubuntu 22.04.3 LTS, release 22.04, codename jammy.

```
root@serveur-lamp:~$ apt update && apt upgrade -y
Atteint:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Atteint:2 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Atteint:3 http://security.ubuntu.com/ubuntu jammy-security InRelease
Lecture des listes de paquets... Fait
Construction de l'arbre des dependances... Fait
Lecture des informations d'etat... Fait
Calcul de la mise a jour... Fait
Les paquets suivants seront mis a jour:
  libssl3 openssl python3-apt
3 mis a jour, 0 nouvellement installes, 0 a enlever et 0 non mis a jour.

root@serveur-lamp:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 22.04.3 LTS
Release:        22.04
Codename:       jammy
```

Figure : Systeme mis a jour - Ubuntu 22.04 LTS

4. Installation d'Apache2

```
sudo apt install apache2 -y
sudo systemctl enable --now apache2
sudo systemctl status apache2
```

```

root@serveur-lamp: ~

root@serveur-lamp:~$ systemctl status apache2
* apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled)
   Active: active (running)
     since Mon 2025-01-15 09:30:12 UTC; 2min ago
    Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 1423 (apache2)
     Tasks: 55 (limit: 2234)
    Memory: 12.4M
       CPU: 142ms
   CGroup: /system.slice/apache2.service
           |-1423 /usr/sbin/apache2 -k start
           |-1424 /usr/sbin/apache2 -k start
           |-1425 /usr/sbin/apache2 -k start

janv. 15 09:30:12 serveur-lamp systemd[1]: Starting The Apache HTTP Server...
janv. 15 09:30:12 serveur-lamp apache2[1421]: AH00558: apache2: Could not reliably...
janv. 15 09:30:12 serveur-lamp systemd[1]: Started The Apache HTTP Server.

```

Figure : Apache2 actif et en cours d'exécution (active running)

```

# Sécurisation : masquer la version Apache dans les en-tetes
sudo nano /etc/apache2/conf-available/security.conf
# Modifier : ServerTokens Prod | ServerSignature Off
sudo systemctl reload apache2

```

5. Installation et sécurisation de MariaDB

5.1 Installation

```

sudo apt install mariadb-server mariadb-client -y
sudo systemctl enable --now mariadb
sudo systemctl status mariadb

```

```

root@serveur-lamp: ~

root@serveur-lamp:~$ systemctl status mariadb
* mariadb.service - MariaDB 10.6.12 database server
   Loaded: loaded (/lib/systemd/system/mariadb.service; enabled)
   Active: active (running)
     since Mon 2025-01-15 09:32:47 UTC; 5min ago
    Docs: man:mariabdb(8)
          https://mariadb.com/kb/en/library/systemd/
   Main PID: 1892 (mariabdb)
  Status: "Taking your SQL requests now..."
     Tasks: 8 (limit: 2234)
    Memory: 74.2M
   CGroup: /system.slice/mariadb.service
           |-1892 /usr/sbin/mariabdb

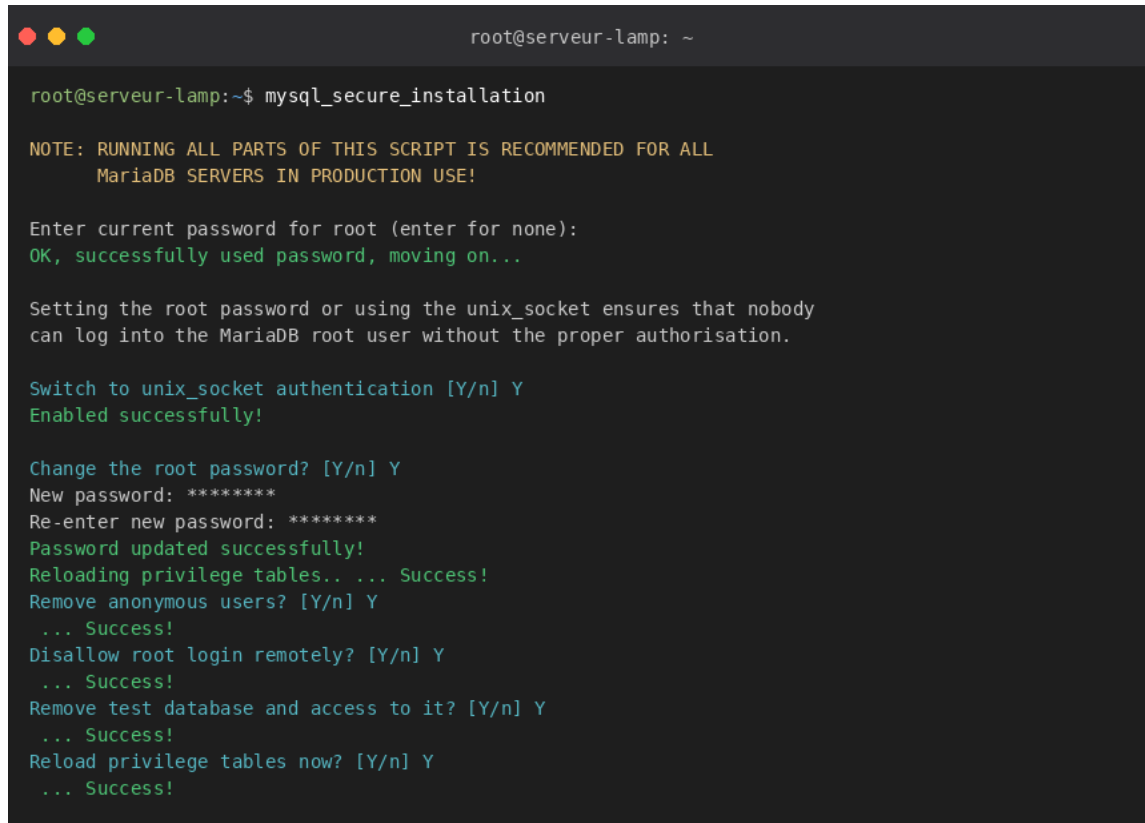
janv. 15 09:32:47 serveur-lamp systemd[1]: Starting MariaDB 10.6.12...
janv. 15 09:32:47 serveur-lamp systemd[1]: Started MariaDB 10.6.12 database server.

```

Figure : MariaDB actif et en cours d'exécution

5.2 Sécurisation avec mysql_secure_installation

```
sudo mysql_secure_installation
```



```
root@serveur-lamp: ~  
  
root@serveur-lamp:~$ mysql_secure_installation  
  
NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL  
      MariaDB SERVERS IN PRODUCTION USE!  
  
Enter current password for root (enter for none):  
OK, successfully used password, moving on...  
  
Setting the root password or using the unix_socket ensures that nobody  
can log into the MariaDB root user without the proper authorisation.  
  
Switch to unix_socket authentication [Y/n] Y  
Enabled successfully!  
  
Change the root password? [Y/n] Y  
New password: *****  
Re-enter new password: *****  
Password updated successfully!  
Reloading privilege tables.. ... Success!  
Remove anonymous users? [Y/n] Y  
... Success!  
Disallow root login remotely? [Y/n] Y  
... Success!  
Remove test database and access to it? [Y/n] Y  
... Success!  
Reload privilege tables now? [Y/n] Y  
... Success!
```

Figure : Sécurisation interactive de MariaDB

5.3 Creation d'une base et d'un utilisateur dedie

```
sudo mysql -u root -p  
CREATE DATABASE db_webapp;  
CREATE USER 'webapp_user'@'localhost' IDENTIFIED BY  
'MotDePasse_Fort123!';  
GRANT ALL PRIVILEGES ON db_webapp.* TO 'webapp_user'@'localhost';  
FLUSH PRIVILEGES;  
EXIT;
```

```

root@serveur-lamp: ~

root@serveur-lamp:~$ mysql -u root -p
Enter password: *****
Welcome to the MariaDB monitor. Commands end with ; or \g.
MariaDB [(none)]>
CREATE DATABASE db_webapp;
Query OK, 1 row affected (0.001 sec)

MariaDB [(none)]>
CREATE USER 'webapp_user'@'localhost' IDENTIFIED BY 'MotDePasse_Fort123!';
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]>
GRANT ALL PRIVILEGES ON db_webapp.* TO 'webapp_user'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]>
FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)
MariaDB [(none)]>
EXIT;
Bye

```

Figure : Base de donnees et utilisateur dedie crees

6. Installation de PHP

```

sudo apt install php libapache2-mod-php php-mysql php-cli -y
php -v
# Test : sudo nano /var/www/html/info.php --> <?php phpinfo(); ?>
# Supprimer apres test : sudo rm /var/www/html/info.php

```

```

root@serveur-lamp: ~

root@serveur-lamp:~$ php -v
PHP 8.1.2-1ubuntu2.14 (cli) (built: Aug 18 2023 11:41:11) (NTS)
Copyright (c) The PHP Group
Zend Engine v4.1.2, Copyright (c) Zend Technologies
    with Zend OPcache v8.1.2-1ubuntu2.14, Copyright (c), by Zend Technologies

root@serveur-lamp:~$ systemctl reload apache2
root@serveur-lamp:~$ curl http://localhost
<!DOCTYPE html PUBLIC ...
<html><body><h1>It works!</h1>
<p>This is the default welcome page used to test the correct
operation of the Apache2 server after installation.</p>
</body></html>

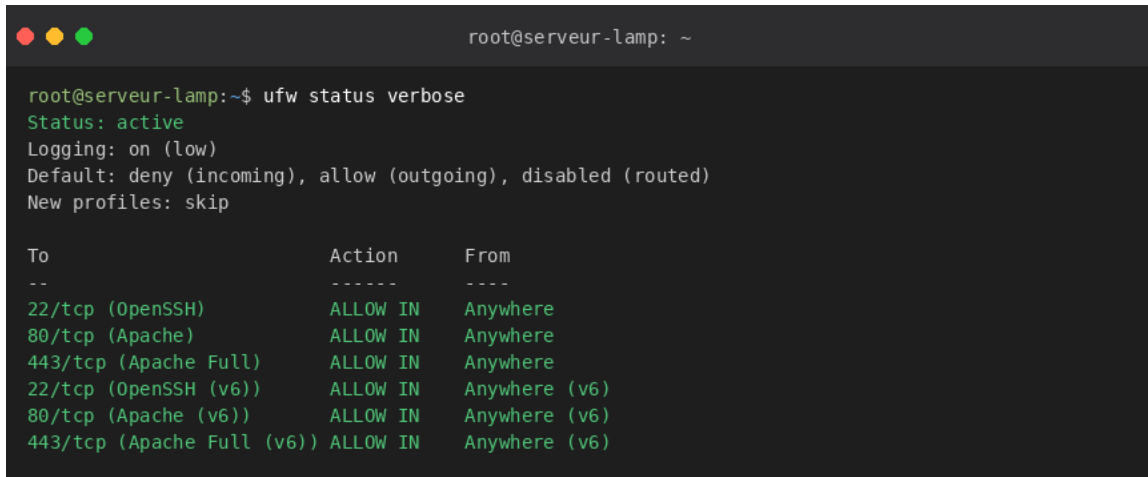
```

Figure : PHP 8.1 installe et integre avec Apache

ATTENTION : Supprimer le fichier info.php immediatement apres verification !

7. Configuration du pare-feu UFW

```
sudo ufw allow OpenSSH
sudo ufw allow 'Apache Full'
sudo ufw enable
sudo ufw status verbose
```



```
root@serveur-lamp: ~
root@serveur-lamp:~$ ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

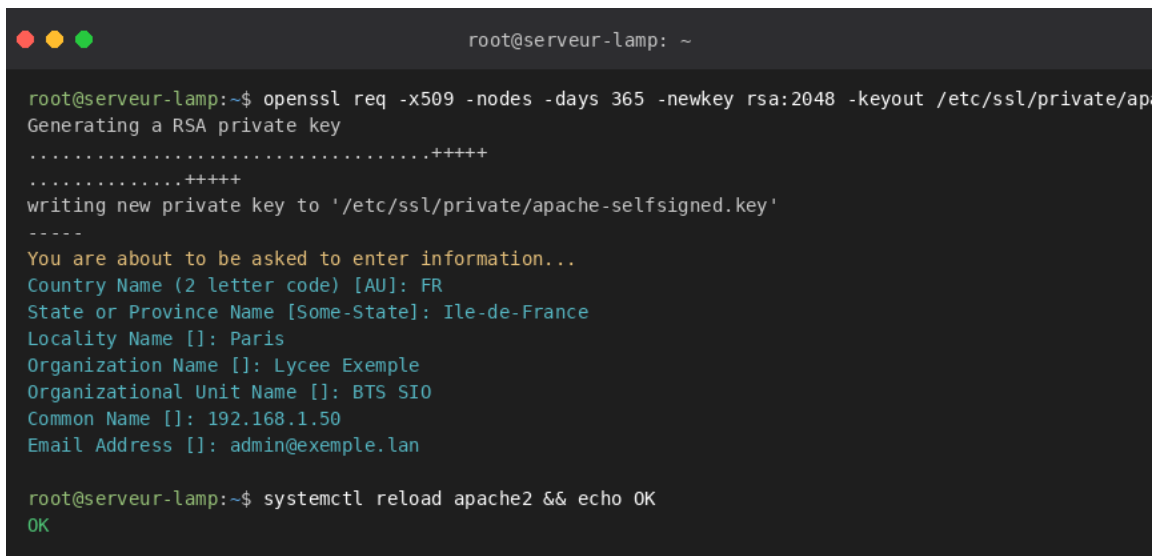
To Action From
--
22/tcp (OpenSSH) ALLOW IN Anywhere
80/tcp (Apache) ALLOW IN Anywhere
443/tcp (Apache Full) ALLOW IN Anywhere
22/tcp (OpenSSH (v6)) ALLOW IN Anywhere (v6)
80/tcp (Apache (v6)) ALLOW IN Anywhere (v6)
443/tcp (Apache Full (v6)) ALLOW IN Anywhere (v6)
```

Figure : UFW actif - regles HTTP (80), HTTPS (443) et SSH (22) ouvertes

INFO : La regle OpenSSH doit être ajoutée AVANT d'activer UFW pour conserver l'accès SSH.

8. Mise en place de HTTPS

```
sudo a2enmod ssl && sudo a2ensite default-ssl
sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
  -keyout /etc/ssl/private/apache-selfsigned.key \
  -out /etc/ssl/certs/apache-selfsigned.crt
sudo systemctl reload apache2
```



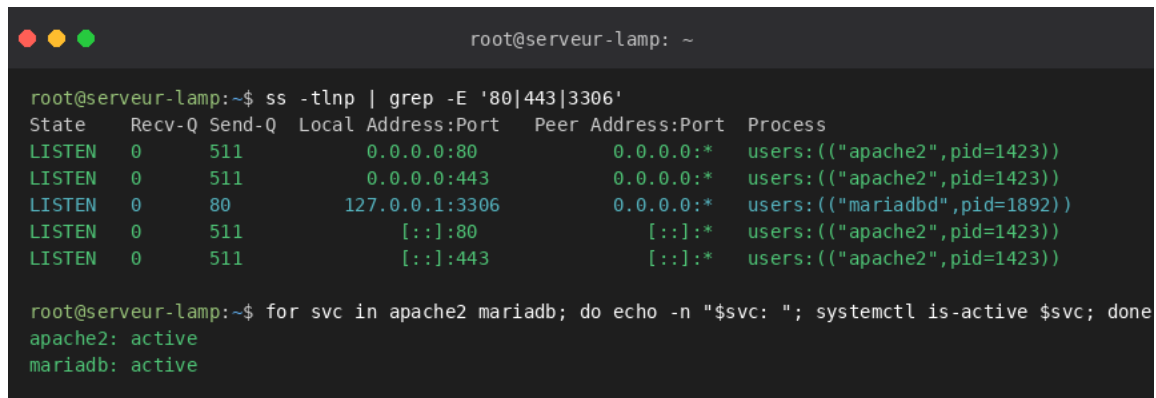
```
root@serveur-lamp: ~
root@serveur-lamp:~$ openssl req -x509 -nodes -days 365 -newkey rsa:2048 -keyout /etc/ssl/private/ap
Generating a RSA private key
.....+++++
.....+++++
writing new private key to '/etc/ssl/private/apache-selfsigned.key'
-----
You are about to be asked to enter information...
Country Name (2 letter code) [AU]: FR
State or Province Name [Some-State]: Ile-de-France
Locality Name []: Paris
Organization Name []: Lycee Exemple
Organizational Unit Name []: BTS SIO
Common Name []: 192.168.1.50
Email Address []: admin@exemple.lan

root@serveur-lamp:~$ systemctl reload apache2 && echo OK
OK
```

Figure : Generation du certificat SSL auto-signé (365 jours, RSA 2048)

9. Verification finale

```
sudo ss -tlnp | grep -E '80|443|3306'
for svc in apache2 mariadb; do echo -n "$svc: "; systemctl is-active $svc; done
```



```
root@serveur-lamp: ~

root@serveur-lamp:~$ ss -tlnp | grep -E '80|443|3306'
State      Recv-Q  Send-Q  Local Address:Port  Peer Address:Port  Process
LISTEN     0        511      0.0.0.0:80          0.0.0.0:*          users: (("apache2",pid=1423))
LISTEN     0        511      0.0.0.0:443        0.0.0.0:*          users: (("apache2",pid=1423))
LISTEN     0        80      127.0.0.1:3306     0.0.0.0:*          users: (("mariadb",pid=1892))
LISTEN     0        511      [::]:80            [::]:*             users: (("apache2",pid=1423))
LISTEN     0        511      [::]:443           [::]:*             users: (("apache2",pid=1423))

root@serveur-lamp:~$ for svc in apache2 mariadb; do echo -n "$svc: "; systemctl is-active $svc; done
apache2: active
mariadb: active
```

Figure : Tous les services en écoute sur les bons ports

9.1 Tableau recapitulatif

Service	Port	Statut	Verification
Apache2	80, 443	active	systemctl status apache2
MariaDB	3306	active	systemctl status mariadb
PHP	N/A	Module Apache	php -v
UFW	N/A	active	ufw status
SSL/HTTPS	443	actif	curl -k https://localhost

9.2 Checklist

- ☐ Apache2 démarre automatiquement
- ☐ Page par défaut Apache accessible en HTTP
- ☐ MariaDB securise (mysql_secure_installation effectuée)
- ☐ PHP fonctionnel
- ☐ UFW actif avec règles HTTP, HTTPS, SSH
- ☐ HTTPS opérationnel
- ☐ Fichier info.php supprime

10. Bilan - Competences acquises

Competence	Acquis	A revoir
Installer et configurer Apache2	☐	☐
Installer et securiser MariaDB	☐	☐
Installer PHP et tester LAMP	☐	☐
Configurer UFW	☐	☐
Mettre en place HTTPS avec OpenSSL	☐	☐
Appliquer les bonnes pratiques de securisation	☐	☐